

Towards Intelligent Recommendations: Fine-Tuning Large Language Models with Collaborative Filtering Techniques

Abstract

Large Language Models (LLMs) have exhibited exceptional performance in multiple domains, leading academics to explore its application in Recommender Systems (RSs). Current research utilizes In-Context Learning (ICL) or supervised fine-tuning on task-specific datasets to tailor LLMs for recommendation tasks in this domain. Nonetheless, whereas LLMs demonstrate adeptness in language understanding, they necessitate further domain-specific knowledge to perform recommendation jobs effectively. Furthermore, when relying solely on textual data, LLMs struggle to effectively capture collaborative signals, which are crucial for accurately aligning recommendations with user preferences and evolving tastes. Due to the need for shared data to understand user preferences, this constraint lowers recommendation quality. This literature review proposes a dual technique for employing LLMs in recommendation contexts to address these issues. This project begins the fine-tuning of LLMs utilizing recommendation data to improve their alignment with recommendation contexts. For improved recommendations and better user alignment, this project utilizes collaborative data. The objective is to develop a model that effectively employs LLMs for recommendation tasks, simultaneously tackling both cold-start and warm-start issues through specialized training and inference methodologies. Furthermore, this methodology significantly improves the model's understanding of collaborative semantics, leading to increased RSs' effectiveness.

Keywords—Recommender Systems, Large Language Model, Collaborative Information

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1 Introduction

Large Language Models (LLMs) like LLaMA [1] and ChatGPT [2] are excellent at tasks that require them to understand semantics, draw conclusions about knowledge, and find information [2]. Previous studies emphasize their significant generalization ability [3], allowing them to integrate and utilize knowledge across diverse scenarios. Additionally, LLMs excel at creating and understanding natural language and advanced knowledge extraction [4]. This makes them essential for compositional reasoning and information-rich applications. Recommender Systems (RSs) are essential for information filtering and delivering users pertinent and customized content [5]. Collaborative Filtering (CF) is a common algorithm for RSs. It models user preferences by discerning patterns in interactions between users and items, making it versatile across applications and proficient in capturing changing preferences [6]. Recent improvements in LLMs offer novel opportunities to improve RSs by integrating LLMs-driven methods. Consequently, research on LLMs for Recommendations (LLM4Rec) is gaining increasing popularity [7].

In-Context Learning (ICL) is the focus of the initial research for LLM4Rec. It is a method that uses natural language prompts to get recommendations from LLMs [8]. However, reliance only on ICL sometimes leads to challenges in producing accurate recommendations. Research indicates ChatGPT either does not respond or predicts positive outcomes (e.g., ‘likes’) [9], as shown in Figure 1. This limitation is likely due to the lack of specific training for recommendation tasks [10]. To address this challenge, significant efforts have been focused on fine-tuning LLMs using relevant recommendation data [11]. Nevertheless, even after fine-tuning, LLMs rely on text tokens and text semantics to represent users and items, which limits their ability to capture collaborative information [12]. Collaborative information, however, is essential in recommendation tasks, particularly those involving rich interactions [13]. Disregarding it can result in suboptimal performance. Therefore, it is important to explore more effective strategies for incorporating LLMs into recommendation tasks.

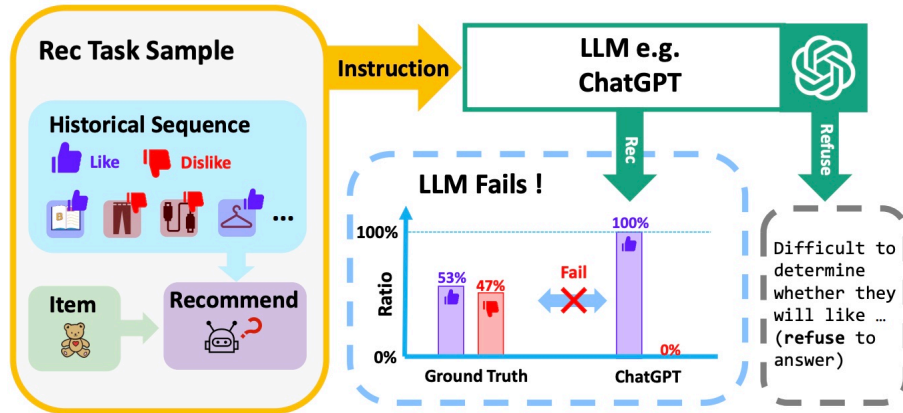


Figure 1: Illustration of ChatGPT refusing to answer or always predicting positively [9].

In summary, this literature review addresses key gaps in the application of LLMs for recommendation tasks by identifying specific limitations in current ICL approaches. This research aims to develop methods that improve recommendation accuracy and adaptability. To achieve this, fine-tuning LLMs for recommendation tasks and integrating collaborative information are proposed to enhance their recommendation capabilities. LLMs will first be fine-tuned using domain-specific data to improve recommendation performance and domain generalization. Since fine-tuning alone cannot effectively capture collaborative information, this methodology incorporates collaborative signals into LLMs to enhance their alignment with user preferences and scalability. A Multi-Layer Perceptron (MLP) is employed to map collaborative data from trained models into the LLMs, ensuring accurate integration for recommendation tasks. Through these efforts, this project seeks to demonstrate the robustness and effectiveness of the proposed method in handling diverse recommendation domains, providing insights into its cross-domain adaptability and practical applications.

2 Literature Review

2.1 Recommender Systems (RSs)

Information overload makes finding relevant content in large catalogs difficult in the digital era. RSs facilitate the efficient retrieval of things of interest, offer a tailored perspective on intricate information landscapes, and prioritize content that corresponds with user preferences [14]. RSs are crucial in different fields such as employment, education, entertainment, and daily life [15], guiding users' decisions and improving information interactions.

RSs frequently face two challenges: the cold-start and warm-start difficulties. The cold-start problem occurs when little interaction data makes it difficult to promote or introduce new things to new or existing consumers. In contrast, the warm-start state is when enough data exists to make effective recommendations based on previous interactions. Warm-start difficulty occurs when RSs have little interaction data for new users or objects, making reliable recommendations difficult [16].

The main ways RSs create recommendations are content-based, collaborative, and hybrid [17]. Collaborative Filtering (CF), as shown in Figure 2, is popular in RSs because it leverages user-item interaction data to find preference patterns. Similarities between people or things based on ratings, clicks, or purchases predict user preferences in CF. This technique predicts future behavior using prior interactions rather than item attributes. CF may be used in many domains and is especially useful in situations with lots of interaction data since it uses interaction data rather than explicit content analysis [18]. CF uses Matrix Factorization (MF) often. The user-item interaction matrix is split into two matrices—one for users and one for items—with each row representing a vector of latent components or embeddings. Factored representations allow MF to forecast missing data, like user ratings, by calculating the inner product of user and item vectors [19].

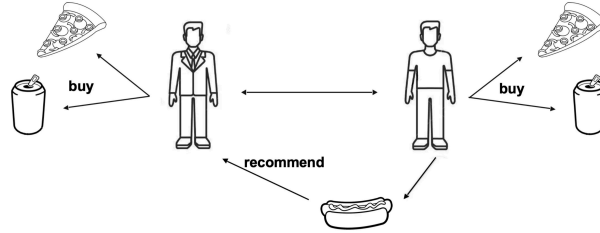


Figure 2: Example of Collaborative Filtering (CF) for Recommendations [20].

Although widely used, CF raises concerns regarding cold-start and data sparsity scenarios [21]. Therefore, a new solution is needed to improve RSs' contextual knowledge, semantic representation, and domain adaptability.

2.2 Large Language Models (LLMs)

LLMs are advanced AI models for complex language processing tasks, excelling in natural language understanding and generation. To handle vast, information-rich inputs efficiently, LLMs employ a structured approach that involves two primary steps, as shown above in Figure 3: **i) Tokenization and Embedding:** The input prompts are broken down into identifiable units or tokens, which are then transformed into high-dimensional vectors through embedding. This process enables the LLMs to map semantic relationships by positioning similar words closer together in the embedding space [22]. LLMs also understand more about semantic and contextual details with the help of transformer architectures and self-attention mechanisms [23]. This improves the quality of embeddings and makes connections between tokens that make sense. **ii) LLMs Prediction:** An MLP, which is a type of feedforward neural network, improves these embeddings even more with a set of linear and nonlinear transformations [24]. This structure simplifies the model's ability to generate accurate responses and supports further processing for response generation [25]. Before reaching the MLP-based layer, LLMs tokenize and embed the inputs.

These features make LLMs valuable tools for applications needing deep semantic and contextual

understanding, especially in representation, reasoning, generalization, and generation. LLMs maintain context and language dependencies, enabling them to adapt to various inputs and provide complicated answers for queries [26]. This skill improves comprehension, disambiguation, and sensitivity to subtleties in text [27]. In both zero-shot and few-shot circumstances, LLMs excel in reasoning and generalization, providing accurate responses with minimal assistance [28]. They effectively share information across domains, leading to more accurate responses in many situations [29]. LLMs leverage extensive world knowledge to deliver context-sensitive answers across various applications. Additionally, they generate interesting, contextually relevant material using individualized datasets to tailor interactions to individual preferences. Dynamic output adjustments based on real-time user inputs enable adaptive and interactive answers. Overall, LLMs are versatile and can provide summaries, recommendations, and explanations to fit varied user demands [30].

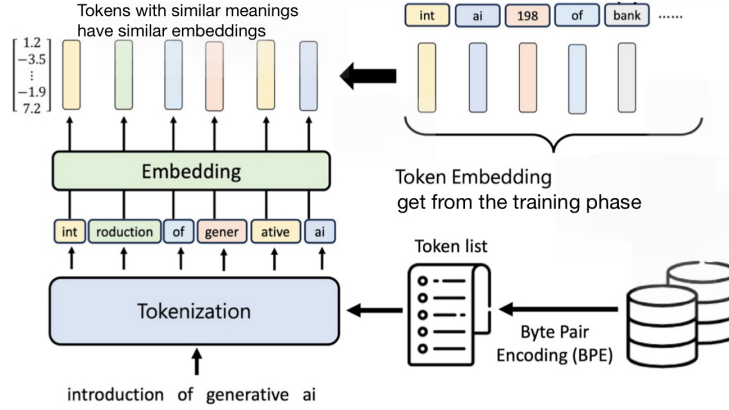


Figure 3: LLMs tokenize and embed inputs before MLP-based prediction refining [31].

LLMs enhance RSs through two key processes: training, where they learn from massive datasets, and inference, when they utilize what they’ve learned to make real-world decisions [32]. The two-phase approach strengthens adaptability, with training to establish a comprehensive knowledge base and inference tailoring outputs to specific, real-time inputs [7]. Within this landscape, Meta’s flexible **LLaMA** stands out among LLMs for its versatility and expertise [33].

2.3 LLMs for Recommendations (LLM4Rec)

The potential of LLM4Rec continues to grow as these models advance. This literature review evaluates RSs, LLMs, and LLM4Rec to assess their suitability for recommendation tasks. Based on existing research, this study identifies gaps and limitations and proposes potential improvements.

Recent research has proposed combining RSs and LLMs to perform recommendation tasks, incorporating the concepts of LLM4Rec. ICL has been proposed as a way for LLMs to adapt to new tasks without fine-tuning by utilizing clues and interaction histories [34]. Thus, LLMs can improve generalization and reduce cold-start problems in CF by incorporating contextual input and modifying recommendations [35]. This strategy, though frequently deemed helpful, may not adequately resolve the challenges LLMs face with extensive recommendation assignments because of insufficient data, substantial computational expenses, word constraints, and efficiency issues [36]. LLMs using ICL lack pre-training on recommendation data, which limits their accuracy. ICL interaction data also records collaborative information through user-item co-occurrence correlations, but LLMs can not get modalities other than text [8]. Since LLMs are trained on generic datasets, this limitation may cause models to exclude target items from candidate lists due to space constraints. Consequently, users may receive fewer personalized recommendations that fail to meet their needs. While LLMs excel at understanding language, they struggle to tailor proposals effectively. Considering these challenges, exploring methods to balance task-specific limitations with the advantages of LLM4Rec could be beneficial.

Past studies have suggested to fine-tuning LLM4Rec. Customizing instructions is essential for enabling LLMs to perform a range of tasks with effective generalization [9]. Based on users preferences

and item usage, model parameters or prompts can be changed. This method improves the module’s recommendation outputs, enabling personalized suggestions and user behavior adaptation [7]. Furthermore, it enhances the relevance and accuracy of recommendations [37], transcending traditional constraints in cold-start settings. However, the focus of this body of work is on cold-start scenarios with scant interaction data, potentially omitting warm-start contexts where collaborative models may find and integrate behavioral patterns. As shown in Figure 4, LLMs face challenges in interpreting collaborative signals when relying solely on textual input, leading to reduced recommendation quality. The absence of collaborative data makes aligning with evolving user interests difficult. Consequently, to improve recommendation accuracy and customization, an alternative perspective could consider integrating collaborative data into LLMs to improve recommendation accuracy and customisation.

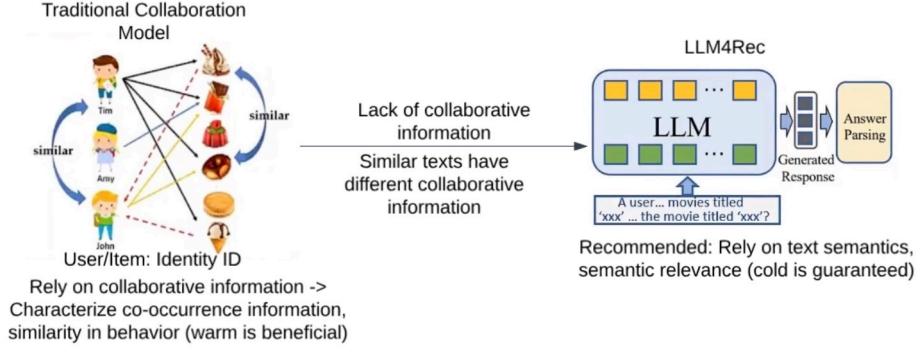


Figure 4: Comparison between Traditional Collaboration Models and LLM4Rec [38].

LLMs struggle to capture the collaborative information essential for recommendations, which hampers their ability to provide high-quality suggestions. To this end, new research suggests using conventional recommendation models (CRMs) in the inference phase of LLMs. This would create a hybrid approach that uses the best parts of both systems [39]. This approach enables more comprehensive recommendations by merging CRMs’ collaborative filtering capabilities with the contextual understanding of LLMs. In this hybrid framework, CRMs and LLMs are initially trained independently. The LLMs integrate CRMs outputs into its processing during a joint inference stage, enabling the LLMs to personalize responses by using CRMs’ interaction data. This structured integration refines recommendations, enhancing their relevance to individual users. Putting together structured user-item data from CRMs with the language understanding of LLMs greatly raises the accuracy of recommendations, leading to more personalized, data-driven RSs [6]. Notably, this method addresses the challenges of warm-start scenarios by utilizing rich user histories, allowing collaborative models to identify and incorporate behavioural patterns accumulated over time.

Together, these studies suggest a broader trend of combining LLMs with CRMs to solve task-specific problems, improve collaborative signal integration, and leverage the power of both models in recommendation scenarios.

3 Project Specification

3.1 Objectives

The potential of LLMs in recommendation contexts serves as the driving force behind this project. This literature review identifies a more comprehensive approach to leveraging the capabilities of LLMs4Rec. To achieve this, the study proposes fine-tuning LLMs for recommendation tasks and explicitly modeling collaborative information within LLMs. The main research goals of this project are outlined below:

Fine-tuning LLMs to enhance their capabilities for advanced recommendation tasks

Although LLMs excel at general language processing, the absence of recommendation-specific data during the pre-training phase impedes their suitability for recommendation tasks. Therefore, this project aims to adapt LLMs for recommendation by synchronizing their functionalities with the

particular requirements of recommendation jobs. During training, LLMs are fine-tuned on recommendation datasets to enhance their adaptability to recommendation contexts and mitigate CRMs cold-start issues. This will improve the model’s recommendation data interpretation and accuracy in low-interaction contexts. Moreover, since this literature review focuses on harnessing the potential of LLMs, this goal is established as the prioritized objective.

Effectively integrate collaborative data into the LLMs to improve performance

From an informational perspective, LLMs encounter challenges in assimilating collaborative data, which is essential for generating accurate recommendations. To address this issue, collaborative data will be directly integrated into LLMs during the inference step. This integration will improve alignment with user preferences and changing tastes, enhancing the model’s performance in warm-start situations where user interaction history exists. Furthermore, integrating recommendation data into the model’s architecture will enable the LLMs to understand collaborative signals more efficiently, improving personalization and the quality of recommendations. This will serve as the subordinate objective of the project.

3.2 Dataset and Baseline

Dataset This study employs the ‘Book-Crossing: User review ratings’ dataset from Kaggle [40], comprising book reviews sourced from Amazon-Book. This dataset comprises 1,149,780 explicit and implicit evaluations for 271,379 books from 278,858 anonymous and demographically categorized individuals. The data will be separated into 8:1:1 training, validation, and testing sets.

Baseline This study will compare LLM-based approaches like ICL on ChatGPT and CRMs like CF-based models like MF to create direct recommendations. To ensure fairness, the project will follow these procedures and guidelines.

3.3 Methodology

The architecture of this project consists of three primary steps: prompt construction, hybrid encoding, and LLMs prediction, as shown in Figure 5. In line with established methodologies, recommendation data is first converted into text-based prompts during the prompt construction phase. These prompts are subsequently encoded and processed through LLMs to generate recommendations during the hybrid encoding and LLMs prediction phases.

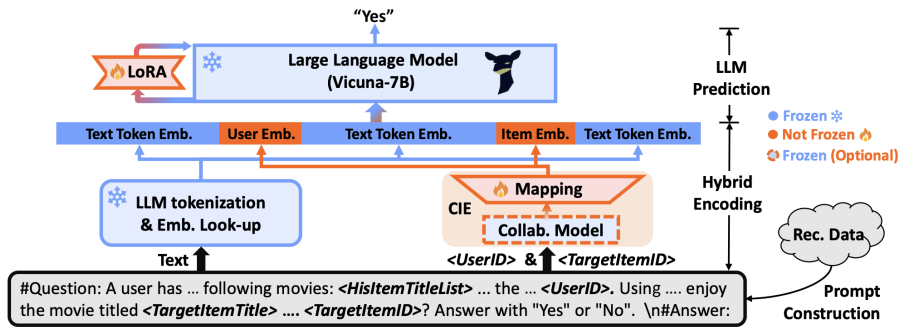


Figure 5: Model Architecture [8].

3.3.1 Prompt Construction

Prior to constructing the initial prompt, the recommendation data is converted into text. This allows the utilization of LLMs’ natural language processing capabilities for recommendation tasks without requiring structural modifications to the data. This approach ensures seamless integration with LLMs. Additionally, fields related to user and item IDs are introduced. These fields are placeholders for collaborative information in the prompt, allowing the model to include contextual data. An example is shown at the bottom of Figure 5.

3.3.2 Hybrid Encoding

In this step, a dual approach is implemented to manage the two data types—text and ID fields—within the structured prompts. First, the LLMs’ built-in tokenization and embedding mechanisms are applied to all textual content, converting the text into tokens and their corresponding embeddings. Next, a CRMs-based Collaborative Information Encoding (CIE) module is used to get collaborative data from the ID fields so that it can be used effectively by the LLMs. After combining the two embeddings, LLMs can make more contextual and collaborative recommendations.

Text Token Embedding The LLMs’ built-in tokenization and embedding mechanisms are utilized to convert all textual content into tokens and their corresponding embeddings, producing text token embeddings. After being tokenized, each token is turned into a continuous vector representation by the LLMs’ internal embedding process. This makes it match up with the language comprehension layer of the model [25]. The tokenization process built in makes sure that input text segments fit perfectly with the model’s training framework, keeping the semantic consistency between tokens and embeddings [41]. By matching tokenized inputs with the model’s language representation, consistent embeddings improve recommendations and task compatibility.

ID Token Embedding ID token embeddings for the “⟨UserID⟩” and “⟨TargetItemID⟩” fields are generated using a CIE module [8]. This module combines MF, a collaborative recommender model, with a mapping layer to encode collaborative data for LLMs. Given a user and an item, the collaborative model generates ID-based representations based on interaction history and user-item patterns. Figure 6 shows how an MLP-implemented mapping layer changes shared representations into the token embedding space of LLMs, which lets LLMs use hidden embeddings.

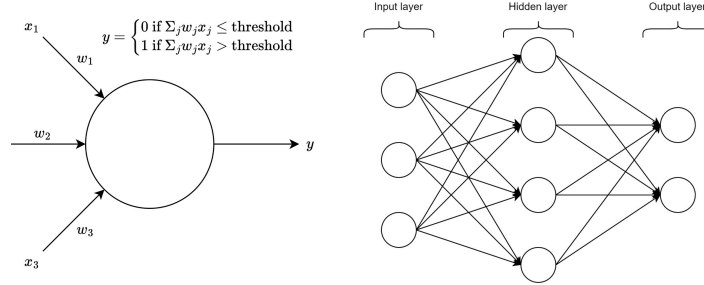


Figure 6: Example of a Perceptron and MLP [42].

Additionally, CIE module training is necessary for precise collaborative data mapping in recommendations. This method constructs prompts from a comprehensive template to minimize module prediction errors. LLMs can utilize collaborative information to enhance the precision of recommendations and contextual relevance through this method.

Embedding Space Mapping Text and ID token embeddings are combined to align them within the token embedding space of the LLMs. This mapping allows the model to integrate semantic content with user and item identifiers, enabling enhanced and contextually relevant recommendations.

3.3.3 Inference Phase

During inference, LLMs leverage the acquired embeddings to generate predictions after data processing and fine-tuning. A LoRA module is employed for recommendation tasks to address the lack of specialized training in LLMs for this purpose. Only the LoRA weights are adjusted to achieve this objective, improve performance, and address the limitations of RSs in cold-start scenarios. Using simply the text-only prompt template and ignoring the marked collaboration data allows parameter-efficient learning for recommendation tasks. This reduces prediction mistakes and improves the module’s task adaptation.

3.4 Deliverables and Milestones

ID	Task	Deliverable	Milestone
1	Prompt Formatting	Prompt JSON Dataset: A structured dataset of recommendation data as LLMs' prompts. Format this dataset according to prompt-based input conventions.	Completion of Prompt Conversion: Converting raw data into structured prompt format, validating LLMs' readiness.
2	Tokenize Text	Tokenized Data: LLMs recommendation text tokenization and embedding. This deliverable should demonstrate LLMs-compatible text token processing.	Completion of Tokenization Process: Complete text tokenization, ensuring the model can use the data with proper embeddings.
3	Fine-Tune LoRA for Recommendations	Fine-Tuned LoRA Module: Customized LoRA module for recommendation tasks. Optimize this module to update only weights for efficiency.	Performance Testing of Fine-Tuned LoRA Module: Finishing LoRA performance testing to ensure it improves accuracy.
4	Build a Mapping Layer	Mapping Layer Module: MLP mapping layer for text and ID embeddings to LLMs' tokens. This deliverable should demonstrate that the MLP matches collaborative data with LLMs' embeddings.	Integration of Mapping Layer: Successfully integrating the MLP mapping layer with the LLMs' embedding space to represent collaborative information.
5	CIE Model Implement	CIE Model Prototype: Functional CIE model employing MF for CF recommendations. The prototype should create collaborative embeddings from user-item interactions.	CIE Model Testing: Initial CIE model testing and validation to enable collaborative embeddings and recommendation data interaction.

3.5 Stretch Goals and Slippage Time

Stretch Goals Cross-domain applicability will be determined by testing the system on movies, literature, and e-commerce. Improve LoRA module latency and memory usage without impacting accuracy.

Slippage Time Prepare extra time to test the system on datasets (cold-start and warm-start scenarios). Spend time perfecting the project report, documentation, and presentation.

3.6 Project Timeline

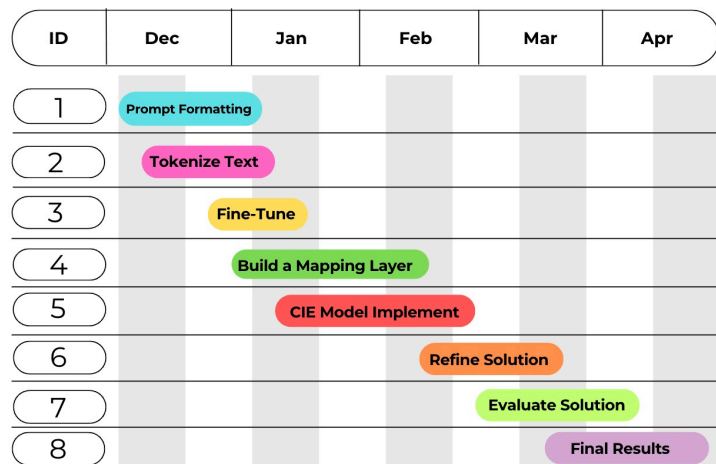


Figure 7: GANTT Chart of the Project Timeline (The prioritized objectives are aligned with the sequence of the task IDs).

4 Evaluations Metrics

ID1: Convert Data to Prompt Format - (Prompt Completeness)

The percentage of recommendation data successfully converted into structured prompts should be at least 95%, ensuring accurate conversion without errors.

ID2: Tokenize Text - (Tokenization Accuracy & Embedding Quality)

The objective is to achieve 99% accuracy in tokenizing text information into LLMs' anticipated format, feasible with proper Python library implementation. For embedding quality, the goal is to ensure a cosine similarity score greater than (>0.9) between embeddings for semantically similar text inputs, reflecting high alignment and consistency in the embedding space

ID3: Fine-Tune LoRA for Recommendations - (Recommendation Performance)

The Area Under the Curve (AUC) statistic will assess recommendation success in predicting user preferences. The AUC measure shows the connection between true positive and false positive rates, indicating forecast accuracy. It measures how successfully the recommendation model prioritizes relevant items over irrelevant ones. A higher AUC score (closer to 1) indicates greater recommendation performance, while 0.5 indicates random guessing [43]. The objective is to surpass a 0.8 AUC score.

ID4: CIE Model Implement - (Precision and Recall for Recommendations)

The purpose is to use precision and recall measures to evaluate embeddings' relevance, with a target score above 80% on datasets. This threshold can be reached by applying normal evaluation methods.

ID5: Build a Mapping Layer - (Mapping Accuracy)

To match collaborative embeddings with LLMs token embedding space, the MLP will be evaluated. The aim is to achieve alignment accuracy above 95% using similarity scores.

5 Ethics, Risks and Mitigation

The dataset for this project relies on vast amounts of user data, including personal preferences and interactions. Significant privacy concerns arise from this reliance, especially when data management is not responsible or secure. Using an Amazon dataset that is available to the public to address these issues guarantees the use of regularization methods, which make data anonymous and ensure strict adherence to data protection laws. Additionally, any data gathered from Amazon has also included user consent, employed encryption, and ensured secure storage to protect user privacy.

Additionally, LLMs and RSs may unintentionally reinforce biases in the training data, unfairly treating particular groups or perpetuating harmful stereotypes. Recommendations may excessively favor particular demographics or material due to skewed historical data. To address these challenges, model results must be regularly checked for bias. Post-processing and fairness algorithms can enhance the equity of recommendations.

6 Conclusions

As LLMs continue to evolve, researchers are increasingly recognizing their potential within RSs. This literature review has examined RSs, LLMs, and LLM4Rec to evaluate how effectively LLMs can be applied to recommendation tasks. Building on current research, it identifies existing gaps and limitations and proposes enhancements to address these challenges.

The proposed project takes a unique approach by combining collaborative data with LLMRec to address CRMs challenges in both warm-start and cold-start scenarios. The goal is to achieve collaborative filtering with pretrained LLMs. To accomplish this, training and inference will be used to externalize LLMs collaborative models. This technique improves collaborative information modeling and mechanism adjustments. Future work will focus on exploring improved methods to leverage LLMs' recommendation capabilities.

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